

Process & Engineering

Process & Engineering group looks after the refinery operations, Gas Processing, LNG terminals and operations of petrochemical plants. Refineries, Gas Processing, LNG and Petrochemical Plants are subjected to high safety hazards from fire & explosion due to handling, processing & storage of highly flammable liquid, gas and vapour as well as operation of the facilities at elevated temperature & pressure or cryogenic condition. These potential hazards get aggravated on accounts of process upsets, extreme physical conditions and accidental release of flammable hydrocarbon.

In recent past, the oil industry throughout its value chain has seen major upgradations and capacity built up in petroleum refining, coupled with induction of newer technologies like Catalytic Hydro-treating, Isomerisation, Hydrocracking, Fluidized Catalytic Cracking, Delayed Coking etc. to extract maximum value addition from the product slate. With these upgradations in technologies along with diversification towards petrochemicals, the complexity of operation has increased manifold and so has our responsibility in operation of the assets in a safe and efficient way without disturbing the people and environment.

Various activities are carried out by this group like 'Safety Audits' (ESA/SSA/PCSA), 'Standards' development and revisioning, investigation of accident/ incidents in related Oil & Gas installations and knowledge dissemination in order to enhance safety in the industry.

The Process & Engineering (P&E) group of OISD while conducting External Safety Audits (ESA), Surprise Safety Audits (SSA) and Pre-Commissioning Safety Audits (PCSA) focusses on the following aspects:

- Design & layout of the installation
- Operation safety and plant operating practices
- Maintenance and inspection aspects
- Safety management and fire protection systems
- Environment compliance
- Training & development activities
- Occupational Health Management

Standards pertaining to P&E section:

SN	OISD-STD/GDN/RP	Standard Name
1	OISD-STD-105	Work Permit System (Statutory as included in The Petroleum Rules 2002)
2	OISD-STD-106	Pressure Relief & Disposal System
3	OISD-STD-108	Oil Storage and Handling (Statutory as included in OMR, 2017)
4	OISD-STD-109	Process Design and Operating Philosophies on Blow Down & sewer system
5	OISD-RP-110	Recommended Practices on Static Electricity
6	OISD-STD-111	Process Design & Operating Philosophies on Fired Process Furnace

7	OISD-STD-112	Safe Handling of Air Hydrocarbon Mixtures & Pyrophoric Substances
8	OISD-STD-113	Classification of Area for Electrical Installations at Hydrocarbon Processing & Handling Facilities
9	OISD-STD-114	Safe Handling of Hazardous Chemicals (Statutory as included in OMR, 2017)
10	OISD-GDN-115	Guidelines on Fire Fighting Equipment and Appliances in Petroleum Industry
11	OISD-STD-116	Fire Protection Facilities for Petroleum Refineries and Oil/Gas Processing Plants (Statutory as included in The Petroleum Rules 2002 & OMR, 2017)
12	OISD-STD-118	Layouts for Oil and Gas Installations (Statutory as included in The Petroleum Rules 2002 & OMR, 2017)
13	OISD-STD-128	Inspection of Unfired Pressure Vessels (Statutory as included in OMR, 2017)
14	OISD-STD-129	Inspection of Storage Tanks (Statutory as included in OMR, 2017)
15	OISD-STD-130	Inspection of Piping Systems
16	OISD-STD-131	Inspection of Boilers
17	OISD-STD-132	Inspection of Pressure Relieving devices
18	OISD-STD-133	Inspection of Fired Heaters
19	OISD-STD-134	Inspection of Heat Exchangers
20	OISD-STD-135	Inspection of Loading & Unloading hoses for Petroleum Products
21	OISD-STD-137	Inspection of Electrical Equipment
22	OISD-STD-141	Cross Country liquid hydrocarbon pipeline (Statutory as included in Petroleum Rules 2002 & OMR, 2017)
23	OISD-STD-142	Inspection of firefighting equipments and systems
24	OISD-STD-144	Liquefied Petroleum Gas (LPG) Installations (Statutory as included in GCR, 2016)
25	OISD-STD-145	Internal Safety Audits
26	OISD-RP-146	Preservation of idle electrical equipment
27	OISD-RP-147	Inspection & safe practices during electrical installations
28	OISD-RP-148	Inspection & safe practices during overhauling electrical equipment
29	OISD-RP-149	Design aspects for safety in electrical systems (Statutory as included in OMR, 2017)
30	OISD-STD-150	Design and Safety Requirements for Liquefied Petroleum Gas Mounded Storage Facility (Statutory as included in SMPV Rules, 2016)
31	OISD-STD-152	Safety Instrumentation for Process System in Hydrocarbon Industry
32	OISD-STD-153	Maintenance & inspection of safety instrumentation in hydrocarbon industry
33	OISD-STD-154	Training in midstream and downstream sector of petroleum industry
34	OISD-STD-155	Personal Protective Equipment
35	OISD-STD-156	Fire Protection Facilities for Ports Handling Hydrocarbons (Statutory as included

		in The Petroleum Rules 2002)
36	OISD-RP-157	Safety in Transportation of Bulk Petroleum Products (Rail and Road)
37	OISD-RP-158	Recommended Practices on Storage and Handling of Bulk Liquefied Petroleum Gas
38	OISD-STD-163	Safety of Control Room for Hydrocarbon industry
39	OISD-STD-164	Fireproofing of Steel Supporting Structures in Oil & Gas Industry
40	OISD-GDN-166	Guidelines for Occupational Health Monitoring in Oil and Gas Industry
41	OISD-STD-170	Inspection, Maintenance, Repairs & Rehabilitation of Foundations & Structures
42	OISD-STD-171	Preservation of Idle Static & Rotary Mechanical Equipment
43	OISD-STD-173	Fire Prevention and Protection System for Electrical Installations
44	OISD-STD-177	Inspection and Maintenance of Insulation System
45	OISD-STD-178	Management of Change (Statutory as included in OMR, 2017)
46	OISD-GDN-180	Lightning Protection
47	OISD-STD-189	Standard On Fire Fighting Equipment for Drilling Rigs, Work Over Rigs and Production Installations
48	OISD-GDN-192	Safety Practices during Construction (Statutory as included in PNGRB)
49	OISD-STD-194	Standard for Unloading, Storage and Regasification of Liquefied Natural Gas (LNG)
50	OISD-GDN-196	Guidelines for Seeking Environmental Clearance of Development Projects in Petroleum Industry
51	OISD-GDN-197	Guidelines for Environmental Impact Assessment
52	OISD-GDN-203	Operation, Maintenance & Inspection of Hoisting Equipment
53	OISD-GDN-206	Guidelines on Safety Management System in Petroleum Industry
54	OISD-GDN-207	Contractor Safety
55	OISD-GDN-211	Safety in Petroleum Laboratories
56	OISD-GDN-224	Monitoring & Control of Volatile Organic Compounds Emission
57	OISD-STD-225	Storage, Handling & Dispensing at Petroleum Retail Outlets
58	OISD-STD-227	Emergency Response and Preparedness in Oil & Gas Industry
59	OISD-GDN-228	Selection, Training & Placement of Fire Operators in Hydrocarbon & Petrochemical Industry
60	OISD-STD-230	Unlined Underground Rock Cavern Storage for Petroleum & Liquefied Petroleum Gas
61	OISD-STD-240	Asset Integrity Management of rotary equipment
62	OISD-STD-241	Safety in the production, storage, and transportation of hydrogen and hydrogen blends
63	OISD-STD-245	Standard on safety for LNG bunkering facilities at ports for large ships, coastal shipping and inland water transport (IWT) terminals

Users are strongly advised to refer to ALL, repeat, ALL OISD Standards / Guidelines / Recommended practice before arriving at any conclusion.